MIRROR: Grounded, Registry-Driven Analytics for Large-Scale Medical Imaging

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***Abstract*—Medical-imaging corpora are now large and heterogeneous: NIH ChestX-ray14 alone is $112{,}120$ frontal radiographs over 14 pathologies, and real studies arrive as DICOM, not as ready-to-train PNGs. MIRROR is a research prototype that treats analytics over such data as a pipeline problem. It ships a real DICOM ingest that applies the modality LUT, the VOI LUT, and the MONOCHROME1 inversion and lifts only non-PHI technical tags, so heterogeneous studies are correctly rendered and routed before any model sees a pixel; and it centralizes each modality's taxonomy, anatomical vocabulary, and report phrasing in one registry, so the same analytics pipeline scales across chest X-ray, brain MRI, and head CT as a data change rather than a re-engineering effort. Its explainability and reporting stages are strictly post-hoc, adding evidence localization and a grounded report for about a 40% wall-clock increase over a bare classifier and changing no prediction. Our central analytics finding is methodological: on the ChestMNIST derivative of ChestX-ray14 the classifier reaches macro AUROC 0.729 and ranks every label 1.6 to 6.8 times better than chance, yet emits no positive prediction for 11 of 14 labels at the default threshold, while a Brier score of 0.045 sits beside the 0.047 a constant predictor earns. Under the class imbalance that dominates large medical corpora, aggregate metrics flatter models that decide nothing; we show that each should be read against the no-skill floor its prevalence implies, which costs one line of arithmetic.**

***Index Terms*—big data analytics, medical imaging, DICOM, data engineering, class imbalance, evaluation methodology, scalability, explainable AI, radiology**

I. Introduction

Deep networks now match specialists on narrow chest-radiograph reading tasks [2], yet adoption remains limited by a trust gap: a bare probability tells a radiologist neither *where* the model is looking nor *why* it concluded what it did, and high-stakes decisions demand models that can be interrogated [16]. Saliency methods such as Grad-CAM [10] expose *where* a prediction comes from, and language models can render structured evidence as prose, but the two are rarely composed into one pipeline whose outputs stay mutually consistent, and whose trust guarantees are stated precisely enough to be checked.

Composing them introduces a second risk the first does not have. A report generator with access to the image can write fluent radiology prose asserting findings no classifier ever produced, and such prose is difficult to audit precisely because it reads well. MIRROR is built around a constraint aimed at that risk: image $\rightarrow$ prediction $\rightarrow$ localization $\rightarrow$ reasoning $\rightarrow$ report, with the language layer receiving only the structured evidence produced upstream, never the raw image. The report can therefore assert only findings the classifier predicted and the localizer situated.

This paper reads MIRROR as an analytics pipeline over large medical-imaging data, and it makes two big-data points that the accuracy of any single classifier does not. The first is about ingestion: turning a hospital-scale DICOM archive into model-ready pixels is real data engineering, because stored pixel values are not display values and a naive read trains a model on images no radiologist would recognize. The second is about measurement: when a corpus is large and heavily imbalanced, the aggregate metrics an analytics dashboard reports will look healthy over a model that makes no decisions at all, and the correction is arithmetic rather than more data. Between the two sits a scaling claim, that one registry lets the same pipeline analyze several imaging modalities as a data change rather than a rewrite.

Research question. Can one grounded pipeline ingest, route, and analyze large-scale medical imaging across modalities, and what does honest measurement of such a pipeline require under the class imbalance that dominates these corpora? We answer the ingestion and scaling questions architecturally, and the measurement question empirically, on chest radiographs. Two things this paper does *not* claim should be stated at the outset. Although MIRROR targets the full 112,120-image NIH release, the reported evaluation is on its downsampled ChestMNIST derivative, chosen for compute, so this is not a demonstration of full-scale training. And it does not measure explanation *quality*: our harness implements a pointing-game and IoU protocol against the NIH lesion boxes, but running it needs a checkpoint trained on the full-resolution release.

Contributions.

1) A data-engineering ingest for heterogeneous medical imaging: a real DICOM reader that applies the modality LUT, VOI LUT, and MONOCHROME1 inversion, lifts only non-PHI tags, and exposes the modality tag so studies are correctly rendered and automatically routed before analytics begins.

2) A registry that lets one analytics pipeline scale across chest X-ray, brain MRI, and head CT by treating each modality's taxonomy, vocabulary, and phrasing as data; three are registered and tested, one is benchmarked.
3) An evaluation-methodology result for imbalanced medical corpora: a model can post strong-looking macro AUROC, Brier, and AUPRC while emitting no positive prediction for 11 of 14 labels, and only reading Brier and AUPRC against their prevalence-derived no-skill floors exposes it.
4) A grounded reporting layer, strictly post-hoc and measured to cost about a 40% wall-clock increase over a bare classifier while changing no prediction, whose language stage receives structured evidence and never the pixels.

II. Related Work

Datasets and classification. Large labeled chest-radiograph corpora made supervised deep learning on this modality practical. The NIH ChestX-ray8/14 release [1] provides 112,120 frontal radiographs over 14 pathologies; CheXpert [3] and MIMIC-CXR [4] extended scale, uncertainty labeling, and paired reports. CheXNet [2] showed a 121-layer DenseNet [6] reaching radiologist-level pneumonia detection. EfficientNet [7] and Vision Transformers [8] complete the standard toolkit; MIRROR keeps all three interchangeable, with DenseNet-121 primary for comparability.

Visual explanation. Class Activation Mapping [9] localized the evidence behind a CNN prediction; Grad-CAM [10] generalized it to arbitrary architectures via gradients, and Score-CAM [11] removed the gradient dependence. Model-agnostic methods, notably LIME [12] and SHAP [13], explain individual predictions without model internals. The pointing-game protocol [14] and intersection-over-union against annotated lesions turn heatmaps into quantitative localization scores; our harness implements both, though we do not run them here.

Critiques of post-hoc explanation. Adebayo et al. [15] showed several saliency methods can be insensitive to the model and data they purport to explain, and argued for sanity checks before any map is trusted. Rudin [16] argued that high-stakes decisions call for inherently interpretable models rather than post-hoc explanations, and Doshi-Velez and Kim [17] called for a measurable science of interpretability. MIRROR is the kind of system these critiques target, and we take up its position against them in Section VII rather than citing them only as background.

Report generation. Systems emitting radiology prose range from TieNet [18] to memory-driven transformers such as R2Gen [19] to decoders optimized for factual correctness [20]. The recurring failure mode is hallucination: prose generated directly from pixels can assert findings the model never detected. Surveys of explainable AI in medical imaging [21] note that interpretability and report generation are usually studied separately. MIRROR's contribution is neither a new classifier nor a new saliency method but their composition under a constraint that bounds what the report may assert.

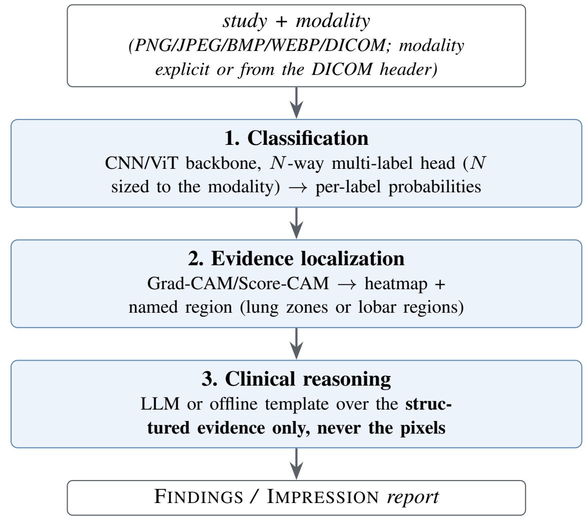


Fig. 1. The MIRROR pipeline as deployed in the local PyTorch stack, the configuration benchmarked in Section V. Layers 2 and 3 are individually toggleable, which recovers the conditions of Section IV. The hosted engine of Section III-F does not have this structure.

III. System Architecture

MIRROR chains three layers; each layer's output is the grounded input to the next (Fig. 1).

A. Layer 1: Classification

The classifier is a factory over three ImageNet-pretrained backbones: DenseNet-121 (default), EfficientNet-B0, and ViT-B/16, with a multi-label head whose width N is the size of the active modality's taxonomy (14 for chest X-ray, 11 each for brain MRI and head CT). The sigmoid is applied at loss and inference time rather than inside the module, so the network emits raw logits. Inputs are resized to the backbone's 224×224 input and normalized with ImageNet statistics; training uses `BCEWithLogitsLoss`, AdamW, a cosine schedule, and dropout 0.2, selecting the checkpoint on best validation macro AUROC.

B. DICOM ingest

Real radiology data arrives as DICOM, and the gap between that and a PNG is not a file-format detail. Stored pixel values are not display values: they must pass through the modality LUT (the rescale slope and intercept that turns raw CT counts into Hounsfield units) and then the VOI or window LUT that selects the diagnostic contrast range, and a `MONOCHROME1` photometric interpretation means the stored values are inverted relative to the usual convention. Skipping any of these silently trains a model on images no radiologist would recognize. MIRROR ships an ingest that applies all three transformations before the pipeline sees a pixel, lifts only non-PHI technical tags, and exposes the modality tag for routing.

C. Layer 2: Evidence Localization

For each positive label (top-k, k=3 by default, to bound compute), this layer computes a class-activation map, either

```
Modality: Chest X-ray
Clinical indication: <text, optional>
<per-modality report guidance>

Structured model evidence (probabilities
from an image classifier, locations from
a Grad-CAM saliency map):
- <Label> (<gloss>): probability=<p>,
  status=PRESENT, localised to <region>
- <Label> (<gloss>): probability=<p>,
  status=below threshold, localised to n/a
  ... one line per label in the taxonomy
```

Fig. 2. The complete input to Layer 3. The reasoning layer sees this text and nothing else: no pixels, no file handle, no image embedding. The grounding property is a consequence of this payload's shape rather than of prompt wording, which is why it can be asserted by a test instead of measured by a benchmark.

Grad-CAM (forward and backward hooks on a per-backbone target layer) or Score-CAM (gradient-free, perturbation-based); ViT maps are recovered by reshaping the patch-token sequence back to a 14×14 spatial grid. The activated region's centroid is mapped into a 3×3 anatomical grid in radiology convention, with patient-right on the viewer's left, and the modality's imaging plane picks the vocabulary: lung zones for a frontal chest film, lobar regions for an axial brain-MRI or head-CT slice. A colormap overlay is rendered for the interface, and the region name, not the heatmap, is what travels downstream.

D. Layer 3: Clinical Reasoning

This layer converts structured evidence into a report with FINDINGS and IMPRESSION sections, using a system prompt plus an evidence-grounded user prompt. Fig. 2 shows the entire payload it receives: one line per label, carrying the label, a plain-English gloss drawn from the registry, the probability, a present or below-threshold status, and the region name from Layer 2. There is no image in the payload and no channel by which one could arrive.

The backend is an LLM (Claude) at low temperature, with a deterministic offline template fallback on any error, so the system yields a coherent report with no network or API key. Predictions below the confidence threshold (0.5) surface as pertinent negatives rather than being dropped, which keeps uncertainty visible. Because both backends consume identical evidence, swapping them changes *how fluently* the report reads, not which findings it asserts. The grounding is therefore at the level of findings, not of the prose framing them, a distinction that carries most of this paper's argument (Section VII).

E. Multi-modality via a modality registry

Swapping a chest radiograph for a brain MRI or head CT changes only the taxonomy the head predicts, the vocabulary a saliency centroid is named in, and a little report phrasing, so a normal brain study reads "no acute intracranial abnormality" rather than "no acute cardiopulmonary abnormality." These live in one registry (Table I), the single source of truth for each modality's labels, glosses, imaging plane, and report guidance;

TABLE I

THE MODALITY REGISTRY. REGISTRATION MEANS ROUTING, HEAD SIZING, AND ANATOMICAL VOCABULARY ARE IMPLEMENTED AND TESTED. ONLY CHEST X-RAY IS TRAINED AND BENCHMARKED HERE.

Modality	**#lbl**	**Taxonomy source**	**Location vocab.**
Chest X-ray	14	NIH ChestX-ray14	lung zones (frontal)
Brain MRI	11	Brain Tumor MRI + neuro findings	lobar regions (axial)
Head CT	11	RSNA ICH subtypes + acute findings	lobar regions (axial)

a hand-kept TypeScript mirror gives the hosted engine and the web interface the identical taxonomy and *ordering*, since output index i maps to labels$[i]$. Modality is resolved from an explicit selection or, for DICOM, from the (0008,0060) tag. Routing, head sizing, anatomical vocabulary, and per-modality phrasing are exercised by the test suite for all three taxonomies, and the post-hoc invariance of Layers 2 and 3 is asserted per modality. No brain-MRI or head-CT checkpoint has been trained, so MIRROR makes no predictive claim on those modalities.

F. Orchestration and serving

A single `analyze()` entry point runs the stages, records per-stage timings, and returns one result object; Layers 2 and 3 are individually toggleable. Two engines satisfy the same JSON contract (Table II), so the web interface (Fig. 3) is identical either way: a finding carries *either* a rendered Grad-CAM overlay from the local stack *or* a normalized bounding box from the hosted one, and the viewer draws whichever is present.

The two engines are not two implementations of one design, and the difference matters for every claim in this paper. The local stack is the architecture of Fig. 1: a trained DenseNet-121, a Grad-CAM map, and a language layer that never receives pixels. The hosted engine exists because that stack does not fit serverless limits, and it replaces the whole pipeline with a single vision LLM under a forced tool call returning the per-label probabilities, one box per present finding, and the report in one response. That engine reads pixels directly, so MIRROR's grounding constraint **does not hold on the hosted path**. Everything benchmarked in Section V is the local stack, and the two hosted-engine figures are labelled as such.

IV. EXPERIMENTAL SETUP

Data. MIRROR targets NIH ChestX-ray14 [1] (112,120 radiographs, 14 multi-label pathologies). Because the full release is $45\,$GB and needs a GPU, we evaluate on **ChestMNIST** [5], its downsampled derivative in MedMNIST v2 (CC BY 4.0), which carries the *identical* 14-label taxonomy in the identical order, so MIRROR runs on it unchanged. We sample $8{,}000$ studies from the pooled official train and validation splits and hold out 10%, giving 7,200 training and 800 validation images, and evaluate on $12{,}000$ test images, a fixed random subsample (seed 42) of the official 22,433-image test split. Training is

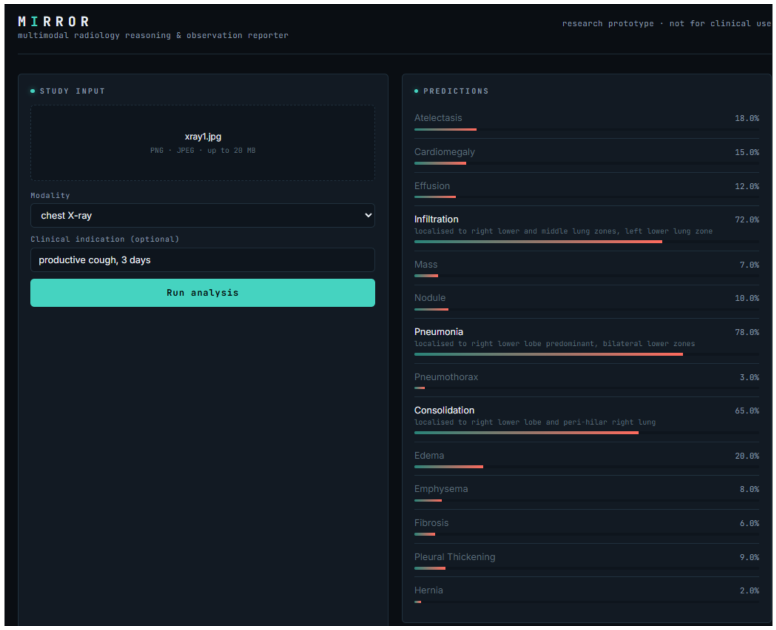
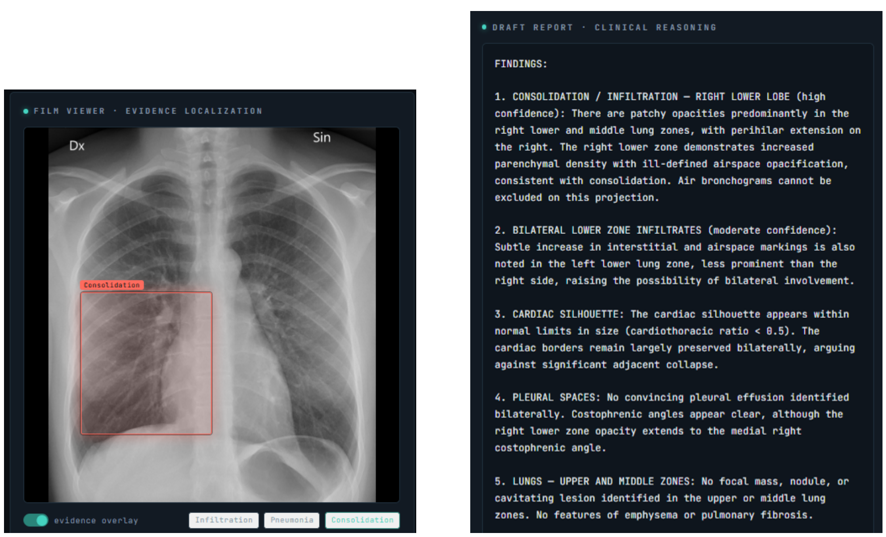


Fig. 3. The MIRROR “reading-room” interface, showing a session on the **hosted serverless engine**, where a vision LLM reads the image directly and returns probabilities, boxes, and report text. This is *not* the benchmarked DenseNet-121 local stack, and no number in this figure is a measured result: it illustrates the interface and the response contract only. A chest radiograph is uploaded with the indication *“productive cough, 3 days,”* all 14 ChestX-ray14 labels are scored, and findings above threshold surface with a saliency-derived location. Both engines drive this identical interface.



(a) Evidence localization

(b) Draft report

Fig. 4. One study end to end, also on the **hosted serverless engine** and not the benchmarked local stack. (a) The localization layer marks the evidence region behind a positive finding and the viewer draws it over the film. (b) The reasoning layer drafts a FINDINGS/IMPRESSION report ending with the mandatory AI-generated-draft disclaimer. The report in (b) is the worked example of Section VIII: alongside its grounded findings it states a cardiothoracic ratio and clear costophrenic angles, neither of which any component of the system measured.

TABLE II
TWO INFERENCE ENGINES BEHIND ONE RESPONSE CONTRACT. ONLY THE LOCAL STACK IMPLEMENTS THE GROUNDED ARCHITECTURE OF FIG. 1, AND ONLY IT IS BENCHMARKED HERE.

	Local stack	**Hosted (serverless)**
Engine	FastAPI + PyTorch	Next.js route, vision LLM
Classify	DenseNet/EffNet/ViT	LLM scores N labels
Localize	Grad-CAM/Score-CAM PNG	LLM returns a bbox
Report	LLM or template	LLM, same call
Sees pixels?	Layer 1 only	Every stage
Inputs	+DICOM, BMP	PNG/JPEG/WEBP
Needs	Python, $\sim$6 GB	one API-key variable

DenseNet-121 on CPU, AdamW at learning rate 3×10^{-4}, weight decay 10^{-5}, batch 32, best of a four-epoch run on validation macro AUROC. Source images are 64×64 and are *upsampled* to the backbone's 224×224 input: there is no genuine 224-pixel information anywhere in this experiment, and findings whose radiographic signature is a few millimetres wide are largely destroyed before training begins.

Metrics and their floors. We report a clinical-reader panel. *Discrimination:* per-label and macro AUROC, and macro AUPRC. *Operating point:* sensitivity, specificity, PPV, and NPV at the 0.5 threshold, the quantities a clinician actually reads off a model. *Calibration:* Brier score and Expected Calibration Error [24] over 10 equal-width bins. Two of these are uninterpretable in isolation on an imbalanced multi-label task, so we compute their no-skill floors from the label prevalences and report both against them: a random ranker's expected average precision equals the prevalence p, and a constant predictor that always returns p scores a Brier of exactly $p(1-p)$. Each headline metric carries a bootstrap 95% confidence interval: the test set is resampled with replacement B=1000 times, with a *single* resample driving all metrics so the intervals are mutually consistent, reported as two-sided percentile intervals (α=0.05).

Post-hoc regression test. Three conditions run the same studies with Layers 2 and 3 switched on and off: classification only, classification with localization, and the full pipeline. Those layers modify no weights and no logits, so the predictions must be identical across conditions; the harness verifies this and profiles the per-stage wall-clock cost of each added layer. This is a test that the implementation matches the design, not an experiment with an uncertain outcome, and Section V-F reports it as such. Every results file stamps a reproducibility record: seed, git commit, and library versions.

V. RESULTS

Every number below is measured on this machine with the local PyTorch stack and traces to a JSON committed in the repository. Table III is the full per-label panel and is the reference for Sections V-A through V-D.

A. Discrimination on ChestMNIST

The DenseNet-121 reaches macro AUROC 0.729 (bootstrap 95% CI $[0.718, 0.738]$) on the 12,000-image test split, with macro AUPRC 0.135 and macro F1 0.031. The published MedMNIST v2 baseline for this dataset, a ResNet-18/50 at 224 px (v2 reports no DenseNet-121), is approximately 0.77 [5]; ours is 0.04 AUROC below it. That ResNet baseline was trained on the full training split for many epochs on a GPU, while ours used 7,200 images for four epochs on a CPU. Both statements are facts and we draw no inference from putting them side by side.

What the per-label ordering does support is that the model learned thoracic structure rather than a dataset artifact (Fig. 5). Readily visible findings rank highest, with Edema at 0.849, Cardiomegaly at 0.830, and Effusion at 0.827, while subtle small ones sit lowest, with Nodule at 0.643 and Infiltration at 0.660. Every confidence interval clears 0.5. The widest interval belongs to Hernia, $[0.615, 0.820]$, which has only 24 positive cases and should be read as the least reliable row in the panel.

B. The operating point: silence on most of the taxonomy

AUROC is threshold-free, and it hides what this model does when asked for a decision. At the default 0.5 threshold, **11 of the 14 labels never produce a single positive prediction** across all 12,000 test studies: their sensitivity is exactly 0.0, their specificity exactly 1.0, and their PPV undefined. Only Effusion, Infiltration, and Pneumothorax fire at all, and Pneumothorax fires on 1.2% of its positive cases. Macro sensitivity is 0.019 and macro F1 is 0.031.

This is not a conservative operating point chosen for a clinical reason. Nothing was tuned: 0.5 is the default, and under prevalences running from 17.5% for Infiltration down to 0.2% for Hernia, a lightly trained model minimizes its loss by pushing almost every output below that line. At this threshold the classifier is not usable as a detector. That is a property of the training budget and the destroyed resolution, not a design decision.

C. Ranking is real where decisions are absent

The natural next question is whether the eleven silent labels are silent because the model learned nothing about them. They are not, and separating those two explanations is the most useful thing the panel does. A uniformly random ranker's expected average precision equals the label's prevalence [22], [23], so AUPRC only means something as a ratio to that floor. Expressed as lift over prevalence, every one of the 14 labels ranks better than chance: from 1.6× for Fibrosis to 6.8× for Cardiomegaly, with a mean of 3.1× (Table III). Cardiomegaly is a clean illustration: prevalence 0.027, AUPRC 0.184, AUROC 0.830 with a confidence interval nowhere near chance, and exactly zero positive predictions.

The model has therefore learned a usable ordering on every label in the taxonomy and converts that ordering into a decision on almost none of them. The failure is located at the threshold and in the calibration of the output scores, not in the representation, which is a more specific diagnosis than "the

TABLE III

THE FULL CLINICAL-READER PANEL ON THE CHESTMNIST TEST SPLIT (DENSENET-121, 64 PX SOURCE UPSAMPLED TO 224 PX, CPU, BEST OF FOUR EPOCHS; N_{TEST}=12,000), SORTED BY AUROC. *Prev.* IS SUPPORT$_{+}/N$. *Lift* IS AUPRC DIVIDED BY PREVALENCE, SO IT IS THE FACTOR BY WHICH THE MODEL BEATS CHANCE AT RANKING THAT LABEL. EVERY LABEL RANKS ABOVE CHANCE; ELEVEN EMIT NO POSITIVE PREDICTION AT THE 0.5 THRESHOLD, WHICH IS WHY THEIR SENSITIVITY IS 0.000, SPECIFICITY 1.000, AND PPV UNDEFINED. HERNIA RESTS ON 24 POSITIVE CASES AND IS THE LEAST RELIABLE ROW. LIFTS USE UNROUNDED AUPRC AND PREVALENCE, SO A ROW'S LIFT CAN DIFFER FROM THE RATIO OF ITS DISPLAYED VALUES. THE **MACRO** LIFT IS THE MEAN OF THE PER-LABEL LIFTS, NOT MACRO AUPRC OVER MACRO PREVALENCE (2.6×); THE MACRO PPV AVERAGES THE ELEVEN UNDEFINED LABELS AS ZERO, AND IS 0.455 OVER THE THREE THAT FIRE.

Pathology	Prev.	AUROC	95% CI	AUPRC	Lift	Sens.	Spec.	PPV
Edema	0.018	0.849	(0.828–0.871)	0.104	5.8×	0.000	1.000	—
Cardiomegaly	0.027	0.830	(0.806–0.851)	0.184	6.8×	0.000	1.000	—
Effusion	0.121	0.827	(0.817–0.838)	0.413	3.4×	0.143	0.987	0.599
Consolidation	0.041	0.754	(0.736–0.773)	0.092	2.2×	0.000	1.000	—
Emphysema	0.023	0.744	(0.713–0.774)	0.069	3.0×	0.000	1.000	—
Pneumothorax	0.049	0.738	(0.719–0.757)	0.138	2.8×	0.012	0.999	0.350
Atelectasis	0.109	0.729	(0.716–0.742)	0.221	2.0×	0.000	1.000	—
Hernia	0.002	0.725	(0.615–0.820)	0.011	5.4×	0.000	1.000	—
Mass	0.049	0.696	(0.673–0.719)	0.125	2.6×	0.000	1.000	—
Fibrosis	0.015	0.674	(0.639–0.705)	0.025	1.6×	0.000	1.000	—
Pneumonia	0.011	0.672	(0.627–0.716)	0.022	2.0×	0.000	1.000	—
Pleural Thickening	0.032	0.666	(0.638–0.692)	0.066	2.1×	0.000	1.000	—
Infiltration	0.175	0.660	(0.648–0.673)	0.298	1.7×	0.110	0.967	0.415
Nodule	0.059	0.643	(0.622–0.662)	0.123	2.1×	0.000	1.000	—
Macro		**0.729**	**(0.718–0.738)**	**0.135**	**3.1×**	**0.019**	**0.997**	**0.097**
Macro F1 0.031	Brier 0.045 against a constant-prevalence floor of 0.047					ECE 0.018		bootstrap B=1000

model underperforms" and points at a different fix: per-label threshold selection on a validation split, not more capacity.

D. Calibration against a no-skill baseline

Read on their own, the calibration numbers look strong: macro Brier 0.045 and ECE 0.018. They are also close to meaningless here, and the arithmetic showing it needs no new experiment. For a label with prevalence p, a constant predictor that ignores the image and always returns p achieves a Brier score of exactly $p(1-p)$. Averaged over the 14 ChestMNIST prevalences, that is a macro Brier of $\mathbf{0.0472}$ for a model that has learned nothing at all. Our trained classifier scores $\mathbf{0.0453}$. The entire measured calibration advantage of a DenseNet-121 over a constant is 0.0019 Brier, about 4%.

That aggregate calibration metrics flatter a model under class imbalance is a documented pitfall, not our finding; the arithmetic above makes it concrete. Under the class imbalance typical of multi-label radiology, where most labels sit below 5% prevalence, $p(1-p)$ is small for almost every label, so the no-skill floor is already low and there is little room between it and a perfect score. A model silent on 11 of 14 labels still posts a Brier of 0.045 and an ECE of 0.018, numbers that would pass unremarked in most results tables. Aggregate calibration metrics on imbalanced multi-label medical tasks should be reported against the constant-prevalence baseline, or not offered as evidence of quality at all.

E. Harness control on synthetic data

Numbers are only worth reporting if the harness that produced them credits real signal and nothing else, so we test that directly. We train the same pipeline on a fully synthetic dataset of 1,400 procedurally generated chest-radiograph-like images whose generator injects a visible blob for seven named labels,

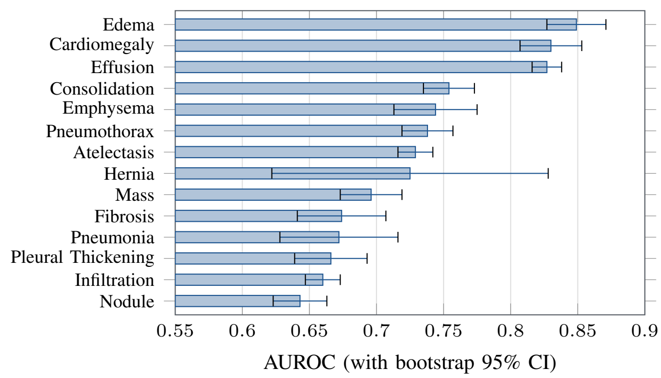


Fig. 5. Per-label test AUROC on ChestMNIST (DenseNet-121, 64 px source upsampled to 224 px, CPU, best of four epochs, N_{test}=12,000), with bootstrap 95% CI whiskers, sorted ascending. The ordering tracks how visible each finding is, and every interval clears the 0.5 chance line. Eleven of these labels nonetheless emit no positive prediction at the default threshold (Section V-B).

Mass, Nodule, Effusion, Infiltration, Consolidation, Edema, and Cardiomegaly, and leaves the other seven with no visual correlate. The seven are fixed in the generator in advance, not chosen after seeing results.

A correct harness should learn exactly the first group and sit near chance on the second, and that is what happens (Fig. 6). On a 350-image held-out synthetic test split the seven signal-bearing labels reach a mean AUROC of 0.917, from Mass at 0.979 down to Edema at 0.866, and the seven no-signal labels average 0.533, from Hernia at 0.620 down to Pleural Thickening at 0.434. The two groups do not overlap: the worst signal-bearing label outscores the best no-signal label by 0.25 AUROC. The no-signal mean sits slightly above 0.5 and straddles chance in both directions, which rules out a

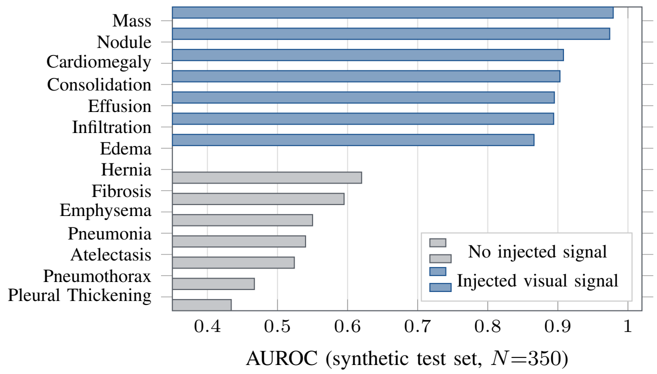


Fig. 6. Harness control on synthetic data. The generator injects a blob for seven labels named in advance and none for the other seven; the classifier learns *exactly* the signal-bearing group (mean AUROC 0.917) and stays near chance on the rest (mean 0.533), with no overlap. This checks that the metrics measure real discrimination rather than label frequency, and that nothing is leaking.

systematic leak. The classifier, the loss, the metrics, and the bootstrap intervals therefore respond to injected signal and not to label frequency or ordering.

F. Post-hoc invariance and latency

Layers 2 and 3 run after the forward pass and modify no weights and no logits, so the predictions are identical across the three conditions by construction. The harness confirms it: the maximum per-label probability change against the classification-only baseline is 0.000 over n=24 studies. This is a regression test, not a result. It earns its place because a nonzero delta would expose a real bug, an accidental coupling between the layers, and the test suite asserts the same property per modality against stubbed backbones so it holds structurally rather than only for the chest checkpoint.

The cost that does vary is latency (Table IV). Prediction takes about 100 ms per study on CPU, the Grad-CAM pass adds 36 to 41 ms, and the offline template report adds 0.03 ms. The two conditions that run Grad-CAM measured 41.4 and 36.2 ms for the same work, which is the run-to-run spread of a 24-study CPU profile and is why the full-pipeline total of 136.2 ms came in below the localization-only total of 140.9 ms. Interpretability here costs roughly a 40% increase in wall-clock time over a bare classifier: all of that sits in the saliency pass, and none in the report.

VI. Data Engineering and Analytics at Scale

The classifier of Section V is a modest one, and it is not the transferable part of this work. What transfers is how a pipeline over large, heterogeneous medical imaging should be built and measured. We collect that here.

Ingestion is the first analytics problem. A hospital-scale imaging archive is DICOM, and the distance between a DICOM file and a trainable pixel array is not a format conversion. Stored pixel values must pass through the modality LUT, the rescale slope and intercept that turns raw CT counts into Hounsfield units, and then the VOI or window LUT that selects

TABLE IV

Ablation conditions with measured per-stage wall-clock latency (ms/study, mean over 24 studies on CPU). Each row is a separate profiling run, so its *Predict* time carries the run-to-run spread of a 24-study CPU profile; within each row *Predict* + *Localize* + *Report* equals *Total*. That spread, not any change in work, is why full MIRROR totals less than localization alone. **There is one AUROC measurement here, not three:** macro AUROC = 0.729 from Table III applies unchanged to every row because the post-hoc layers cannot alter a prediction.

Condition	Predict	Localize	Report	Total
Classification only	101.2	—	—	101.2
+ Evidence localization	99.5	41.4	—	140.9
Full MIRROR	100.0	36.2	0.03	136.2

Macro AUROC = 0.729 in every row: one measurement.
Max probability change vs. baseline = 0.000, n=24.

the diagnostic contrast range; a `MONOCHROME1` photometric interpretation inverts the stored values relative to the usual convention. Skip any of these at scale and the analytics run silently on images no radiologist would recognize, most visibly the `MONOCHROME1` studies, which arrive as photographic negatives. MIRROR's ingest applies all three transformations before the pipeline sees a pixel, lifts only non-PHI technical tags, and exposes the modality tag so that ingestion and routing are the same pass. This is the unglamorous data engineering that a correctness claim on a large corpus rests on.

Scaling across modalities is a data change. A large medical-imaging program is rarely one modality. MIRROR centralizes the only things that differ between chest X-ray, brain MRI, and head CT, the taxonomy, the anatomical vocabulary, and a little report phrasing, in one registry (Table I), so the same ingest, backbone factory, localizer, and report structure serve all three. Adding a fourth modality is a registry entry, not a pipeline fork, and a hand-kept TypeScript mirror keeps the hosted engine and the interface on the identical taxonomy and ordering. Three taxonomies are registered, routed, and tested; one is trained.

Throughput is an analytics cost. At corpus scale the per-study cost of the explainability and reporting stages is a budget line. MIRROR's stages are strictly post-hoc: they change no prediction (Section V-F), and their measured cost is about 40% wall-clock over a bare classifier, essentially all of it in the Grad-CAM pass and effectively none in the report (Table IV). An operator can therefore add or remove evidence localization and the grounded report per run, trading interpretability against throughput, with a known and modest price.

At scale, aggregate metrics mislead, and the fix is arithmetic. This is the methodological core, and it matters more the larger and more imbalanced the corpus. Under the prevalences of a real thoracic corpus, most labels sit below 5%, so two of the metrics an analytics report leans on are uninterpretable in isolation. A uniformly random ranker's expected average precision equals a label's prevalence [22], [23], and a constant predictor that always returns the prevalence p scores a Brier of exactly $p(1-p)$; both floors are small when p

is small, leaving almost no room between them and a perfect score. On ChestMNIST our classifier posts macro Brier 0.045 against a constant-prevalence floor of 0.047, an advantage of about 4%, and macro AUROC 0.729, while emitting no positive prediction for 11 of 14 labels. Read against their floors the same numbers tell the true story: AUPRC as lift over prevalence is $1.6\times$ to $6.8\times$, so the ranking is genuinely informative on every label, and the failure is at the threshold, not in the representation. Any analytics pipeline reporting Brier, AUPRC, or macro AUROC over an imbalanced medical corpus should print the no-skill floor beside each, or not offer the metric as evidence of quality. It costs one line of arithmetic and changes how the dashboard reads.

VII. Discussion

Finding-level grounding is a real guarantee and a narrow one. Because the language layer receives a label, a probability, and a region name rather than an image (Fig. 2), the set of findings the report may assert is exactly the set the classifier produced. That is checkable without trusting anything about the model: a reader holding the report and the probability vector can verify the correspondence, and a unit test can assert it. It is also strictly a claim about *which findings appear*, and says nothing about the sentences around them. A report that correctly asserts consolidation may frame it with clauses the system never computed, and those clauses inherit the credibility of the finding they sit beside (Section VIII). Architectures that constrain only the former should say so rather than claiming groundedness in general.

Where this stands against Rudin. MIRROR is precisely the architecture Rudin [16] argues against for high-stakes decisions: an opaque CNN with a saliency map and a language layer attached after the fact. We have no rebuttal, and we do not claim the saliency map is faithful; the sanity checks of Adebayo et al. [15] are the right instrument and we have not run them. What we claim is narrower and does not depend on the heatmap being faithful: the finding set is auditable, because a reader can check the report against the probability vector directly. That is a retreat from explanation to auditability, and we would rather name it than dress it up. An inherently interpretable classifier would be the stronger answer.

Discrimination and decisions are different failures. The point behind Section VI is worth stating conceptually, because it is what makes the floor arithmetic necessary rather than pedantic. Ranking quality and decision quality are separate properties, and the usual headline metrics report only the first. A model can hold a genuinely informative ordering on every label yet convert it into a positive decision on almost none, and no single aggregate number reveals that split: AUROC alone hides the missing decisions, and the operating point alone hides that the representation is sound and the fix is a per-label threshold rather than a bigger network. For any analytics program over medical imaging the practical rule is to publish sensitivity and PPV alongside every AUROC, and to write AUPRC and Brier against their prevalence-derived floors.

One contract, two very different engines. Serving the same JSON response from a PyTorch stack and from a hosted vision LLM shows the *interface* MIRROR proposes, namely predict, show evidence, explain, is independent of the inference backend, at the price of the asymmetry in Section III-F: the grounding guarantee holds only on the local stack, and we flag that rather than let one system's architecture diagram vouch for the other's behaviour.

VIII. Ethics, Safety, and Limitations

Not a medical device. MIRROR is a research prototype, not reviewed or cleared by any regulatory body; every output is a draft that must be verified by a licensed radiologist and must not be used for diagnosis or treatment. Each report carries that disclaimer explicitly.

A worked example of ungrounded detail. Fig. 4 shows a generated report beside its evidence overlay. Two phrases in it are worth quoting: the cardiac silhouette is described as normal *with a cardiothoracic ratio below* 0.5, and the *costophrenic angles are noted to be clear*. The system measured neither. No cardiothoracic ratio was computed, no cardiac or thoracic boundary was segmented, and no costophrenic angle was localized; the pipeline produced a probability per label and up to three saliency centroids, and nothing in that evidence supports either clause. Both are fluent radiology register generated to fill out a report. They also sit in the same paragraph as assertions that *are* grounded, and that adjacency is the danger: a reader who checks the consolidation finding against the overlay and finds it correct has been given a reason to extend trust to the sentence beside it. Fluency makes ungrounded detail more dangerous, not less, because it arrives wearing the credibility of the verified finding next to it. This is why finding-level grounding cannot be described as making a report safe, and why the AI-generated-draft disclaimer is load-bearing rather than boilerplate.

Human subjects and data governance. No ethics-board review was required: this work uses only existing, public, de-identified benchmark datasets, namely NIH ChestX-ray14 and its MedMNIST derivative, and collects no new data from human subjects. Those datasets are governed by their own data use agreements; no raw images, weights, or protected health information are committed to version control, and the DICOM ingest extracts only non-PHI technical tags.

Limitations. Our one quantitative benchmark is ChestMNIST, at 64-pixel source resolution on a 7,200-image budget, and at its default threshold the classifier is not a working detector (Section V-B). No clinical claim of any kind follows from it. Multi-modality is architectural: the brain-MRI and head-CT paths are implemented and exercised by the test suite but have no trained checkpoints. Explanation quality is unmeasured: the pointing-game and IoU protocol is implemented but not run, and no reader study has been conducted, so we have no evidence that the localization or report layers help a human, only that they cost about $40\,\mathrm{ms}$. Report quality is assessed only qualitatively; a comparison against real radiologist reports is future work.

IX. CONCLUSION

MIRROR treats analytics over large medical imaging as a pipeline problem: a real DICOM ingest that renders and routes heterogeneous studies correctly, a registry that scales the same pipeline across chest X-ray, brain MRI, and head CT as a data change, and strictly post-hoc explainability and reporting that cost about 40% wall-clock and change no prediction. Its most portable result is methodological. On the ChestMNIST derivative the classifier reaches macro AUROC 0.729, ranks every label between 1.6 and 6.8 times better than a random ranker, and still emits no positive prediction for 11 of its 14 labels while posting a Brier score within 0.002 of a predictor that never looks at the image. Under the class imbalance that dominates large medical corpora, aggregate metrics can look healthy over a model that makes no decisions, so an analytics pipeline should state each number against the no-skill floor its prevalence implies. Next steps are full-resolution training on the NIH release, per-label threshold selection on a validation split, trained checkpoints for the brain-MRI and head-CT paths, the saliency sanity checks we cite but did not run, and a reader study.

REPRODUCIBILITY

Code: https://github.com/vignesh-nagarajan-vn/MIRROR. Live demo, on the hosted engine: https://mirror-ten-jet.vercel.app/. A zero-setup CLI demo (`python -m demo.run_demo <image>`) runs the local pipeline on ImageNet weights with the offline template backend, needing no checkpoint and no API key. Every results JSON stamps a `reproducibility` block with the seed, git commit, and library versions, and the unit tests are torch-free. The ChestMNIST result (`configs/chestmnist.yaml`) and the synthetic control (`configs/synthetic.yaml`) regenerate from seed 42; their outputs are committed under `results/`. The prevalences, the AUPRC lifts, the no-skill Brier floor, and the exact contents of Table III are recomputed from those files by `paper/calibration_baseline.py`.

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REFERENCES

[1] X. Wang, Y. Peng, L. Lu, Z. Lu, M. Bagheri, and R. M. Summers, "ChestX-ray8: Hospital-scale chest X-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases," *CVPR*, pp. 2097–2106, 2017.

[2] P. Rajpurkar, J. Irvin, K. Zhu, *et al.*, "CheXNet: Radiologist-level pneumonia detection on chest X-rays with deep learning," *arXiv:1711.05225*, 2017.

[3] J. Irvin, P. Rajpurkar, M. Ko, *et al.*, "CheXpert: A large chest radiograph dataset with uncertainty labels and expert comparison," *AAAI*, 2019.

[4] A. E. W. Johnson, T. J. Pollard, S. J. Berkowitz, *et al.*, "MIMIC-CXR, a de-identified publicly available database of chest radiographs with free-text reports," *Scientific Data*, vol. 6, no. 1, p. 317, 2019.

[5] J. Yang, R. Shi, D. Wei, Z. Liu, L. Zhao, B. Ke, H. Pfister, and B. Ni, "MedMNIST v2: A large-scale lightweight benchmark for 2D and 3D biomedical image classification," *Scientific Data*, vol. 10, no. 1, p. 41, 2023.

[6] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," *CVPR*, pp. 4700–4708, 2017.

[7] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," *ICML*, 2019.

[8] A. Dosovitskiy, L. Beyer, A. Kolesnikov, *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," *ICLR*, 2021.

[9] B. Zhou, A. Khosla, A. Lapedriza, A. Oliva, and A. Torralba, "Learning deep features for discriminative localization," *CVPR*, pp. 2921–2929, 2016.

[10] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," *ICCV*, pp. 618–626, 2017.

[11] H. Wang, Z. Wang, M. Du, *et al.*, "Score-CAM: Score-weighted visual explanations for convolutional neural networks," *CVPR Workshops*, 2020.

[12] M. T. Ribeiro, S. Singh, and C. Guestrin, "'Why should I trust you?': Explaining the predictions of any classifier," *KDD*, pp. 1135–1144, 2016.

[13] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," *NeurIPS*, 2017.

[14] J. Zhang, S. A. Bargal, Z. Lin, J. Brandt, X. Shen, and S. Sclaroff, "Top-down neural attention by excitation backprop," *ECCV*, 2016.

[15] J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim, "Sanity checks for saliency maps," *NeurIPS*, 2018.

[16] C. Rudin, "Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead," *Nature Machine Intelligence*, vol. 1, no. 5, pp. 206–215, 2019.

[17] F. Doshi-Velez and B. Kim, "Towards a rigorous science of interpretable machine learning," *arXiv:1702.08608*, 2017.

[18] X. Wang, Y. Peng, L. Lu, Z. Lu, and R. M. Summers, "TieNet: Text-image embedding network for common thorax disease classification and reporting in chest X-rays," *CVPR*, 2018.

[19] Z. Chen, Y. Song, T.-H. Chang, and X. Wan, "Generating radiology reports via memory-driven transformer," *EMNLP*, 2020.

[20] G. Liu, T.-M. H. Hsu, M. McDermott, *et al.*, "Clinically accurate chest X-ray report generation," *MLHC*, 2019.

[21] E. Tjoa and C. Guan, "A survey on explainable artificial intelligence (XAI): Toward medical XAI," *IEEE TNNLS*, vol. 32, no. 11, pp. 4793–4813, 2021.

[22] J. Davis and M. Goadrich, "The relationship between precision-recall and ROC curves," *ICML*, pp. 233–240, 2006.

[23] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," *PLOS ONE*, vol. 10, no. 3, p. e0118432, 2015.

[24] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," *ICML*, pp. 1321–1330, 2017.